

Total recombination and energy-loss coefficients for hydrogenic ions at low density for $10 \leq T_e/Z^2 \leq 10^7$ K

D. G. Hummer^{1,2}

¹Max-Planck-Institut für Astrophysik, Postfach 1523, 85740 Garching bei München, Germany

²Institut für Astronomie und Astrophysik der Universität München, Scheinerstrasse 1, 81679 München, Germany

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ABSTRACT

Total recombination and energy-loss coefficients, including the free-free component, are calculated for Cases A and B from exact hydrogenic photoionization and free-free cross-sections, and are tabulated for $\log(T_e/Z^2) = 1(0.2)7$.

Key words: atomic data.

1 INTRODUCTION

The description of the ionization structure of low-density material surrounding a radiating object, developed by Strömberg (1939) and by Menzel and co-workers in a well-known series of papers in the *Astrophysical Journal*, has proven to be one of the most durable techniques in astrophysical theory. Although the distinction between Case A and Case B introduced by Baker & Menzel (1938) is not completely sharp, it remains a very useful concept. The breakdown of the Case A–B distinction and the generalization to intermediate situations has been treated extensively by Hummer & Storey (1987, 1992).

Although recombination theory was originally applied to objects with temperatures of order 10^4 K, subsequent applications have involved electron temperatures from a few hundred kelvin in the interstellar medium to 10^7 K in Ly α clouds. It appears, however, that no systematic and accurate calculation of total recombination and energy-loss coefficients for hydrogenic ions has been made for the upper end of this temperature range.

The total recombination coefficients $\alpha_A(T_e, Z)$ and $\alpha_B(T_e, Z)$ for Cases A and B, respectively, which relate the recombination rate of hydrogenic ions of charge Z to the product of the ion and electron densities, are in principle calculable to any desired accuracy. Many authors have concerned themselves with the calculation of the recombination coefficients α_n to the n th quantum level of a hydrogenic ion using an asymptotic expansion of the b-f Gaunt factor (Seaton 1959; Burgess 1965) and exact photoionization cross-sections (Boardman 1964), together with the detailed balance relation of Milne (1924). However, it appears that the only source of zero-density *total* recombination coefficients and the corresponding energy-loss coefficients is the work of Seaton (1959), who gives formulae and tables for the

coefficients in expansions in powers of $\lambda^{-1/3}$, where $\lambda = Z^2 \text{ Ryd}/T_e$. Total recombination coefficients for a large range of non-zero densities, including the effects of collisions, and for temperatures in the neighbourhood of 10^3 – 10^4 K, have been obtained from solutions of the equation of statistical equilibrium by Hummer & Storey (1987) and Storey & Hummer (1988, 1994).

We use the exact hydrogenic photoionization cross-sections to evaluate α_n and β_n for values of n as large as 800, and an accurate free-free Gaunt factor to calculate β_{ff} , for temperatures between 10 and 10^7 K.

2 BASIC EQUATIONS

The rate of recombination to level n of a hydrogenic ion of charge Z in a gas with electron temperature T_e can be written as $N_e N(Z) \alpha_n(T_e, Z) \text{ cm}^{-3} \text{ s}^{-1}$, and the corresponding loss of electron energy as $N_e N(Z) k T_e \beta_n(T_e, Z) \text{ erg cm}^{-3} \text{ s}^{-1}$. The recombination and energy-loss coefficients can be expressed compactly as

$$\alpha_n(T_e, Z) = \frac{c\alpha^3}{\sqrt{\pi}} Z^3 \lambda^{3/2} n^{-2} \int_0^\infty d\epsilon (1 + n^2 \epsilon)^2 e^{-\lambda \epsilon} \tilde{a}_n(Z, \epsilon) \quad (1)$$

and

$$\beta_n(T_e, Z) = \frac{c\alpha^3}{\sqrt{\pi}} Z^3 \lambda^{5/2} n^{-2} \int_0^\infty d\epsilon \epsilon (1 + n^2 \epsilon)^2 e^{-\lambda \epsilon} \tilde{a}_n(Z, \epsilon), \quad (2)$$

where $\tilde{a}_n(Z, \epsilon)$ is the photoionization cross-section for level n , ϵ is the energy of the ejected (recombining) electron in $Z^2 \text{ Ryd}$ and

$$\lambda = \frac{Z^2 \text{ Ryd}}{T_e} = 1.5789 \times 10^5 \frac{Z^2}{T_e}. \quad (3)$$

The numerical value of the constant is

$$\frac{c\alpha^3}{\sqrt{\pi}} = 6572.6617 \text{ cm s}^{-1}. \quad (4)$$

It follows directly from equations (1) and (2) that

$$\lim_{T_e \rightarrow 0} \lambda^{1/2} \alpha_n(T_e, Z) = \lim_{T_e \rightarrow 0} \lambda^{1/2} \beta_n(T_e, Z) = \frac{c\alpha^3}{\sqrt{\pi}} Z^3 n^{-2} \tilde{a}_n(0, Z). \quad (5)$$

The free-free energy-loss coefficient can be written as

$$\beta_{ff}(T_e, Z) = \frac{8}{3\pi^2} \left(\frac{\pi}{3}\right)^{1/2} \alpha^2 \lambda_c^2 c Z \lambda^{1/2} \int_0^\infty du e^{-u} g_{III}(u, \lambda), \quad (6)$$

where $u = h\nu/kT_e$, α is the fine-structure constant, λ_c is the Compton wavelength and g_{III} is the free-free Gaunt factor. The numerical value of the constant is

$$\frac{8}{3\pi^2} \left(\frac{\pi}{3}\right)^{1/2} \alpha^2 \lambda_c^2 c = 2.598\,5262 \times 10^{-14} \text{ cm}^3 \text{ s}^{-1}. \quad (7)$$

3 NUMERICAL EVALUATION

The integrals in equations (1), (2) and (6) can be evaluated numerically using the exact values of $\tilde{a}_n$ and the accurate values of g_{III} available from the subroutines published by Storey & Hummer (1992). The photoionization cross-sections are generated by a stable recursion procedure, and the free-free Gaunt factors are computed from a two-dimensional Chebyshev expansion developed by Hummer (1988). [Table 3 of this reference also contains the same information as the celebrated graphs given by Karzas & Latter (1961), and is much easier to use.]

As the photoionization cross-sections in equations (1) and (2) are approximately proportional to $(1+n^2\epsilon)^{-3}$, the integrands were written in the form

$$\int_0^\infty d\epsilon \epsilon^{0 \text{ or } 1} (1+n^2\epsilon)^{-1} e^{-\lambda\epsilon} [\tilde{a}_n(Z, \epsilon)(1+n^2\epsilon)^3]. \quad (8)$$

The function in square brackets varies smoothly with ϵ , and could be approximated well by linear splines. The epsilon integrations could then be evaluated analytically for each interval $[\epsilon_m, \epsilon_{m+1}]$. The discrete values of ϵ were chosen so that variations of the cross-section on the scale of $\delta\epsilon = n^{-2}$ could be accurately represented. Once $\epsilon_m n^2$ was of order unity, the interval lengths were slowly increased. The integrations terminated when $\lambda\epsilon_m$ reached approximately 20. The values of a_1 and a_{10} for $\log T_e = 3(1)6$ agreed with those of Boardman (1964) to one digit in the fourth figure. The values of the individual α_n and β_n are thought to be accurate to more than 4 significant figures, as estimated by the convergence as the quadrature grid was refined.

In evaluating the integral in equation (6), the logarithmic variation of $g_{III}(u, \lambda)$ with u must be taken into account. The value of the integral from $u=0$ to $u=10^{-4}$, the smallest values for which the Gaunt factor is available from the subroutines of Storey & Hummer (1992), is easily shown to be completely negligible.

The evaluation of the sum over n is the major computational problem in this work, as it converges very slowly for low temperatures (and not at all for $T_e=0$). The behaviour of these summations can be seen from Fig. 1, where the logarithm of the ratio $\alpha_n(T_e, Z)/\alpha_1(T_e, Z)$ is plotted versus $\log n$ for $\log T_e = 1(1)7$. It can be seen that, for sufficiently large n , $\alpha_n \sim n^{-3}$. Therefore, if n lies in this asymptotic regime, the sum over higher terms is

$$\begin{aligned} \Delta(n) &\equiv \sum_{n'=n+1}^\infty \alpha_{n'} \approx \alpha_n n^3 \sum_{n'=n+1}^\infty n'^{-3} \\ &= \alpha_n n^3 \left[\frac{1}{2} \left(\frac{1}{n+1} \right)^3 + \int_{n+1}^\infty dn'/n'^3 \right] \\ &= \alpha_n \left(\frac{n}{n+1} \right)^3 \left(1 + \frac{1}{2} n \right), \end{aligned} \quad (9)$$

where the Euler-Maclaurin formula was used to approximate the summation. This correction could easily be verified at high temperatures, and was therefore used at all temperatures. At the lowest temperatures, the calculations had to be carried to $n=800$ before the correction was applied. In any event, the correction was in no case larger than a few tenths

Table 1. Recombination and energy-loss coefficients multiplied by $T_e^{1/2}$.

$\log T_e$	$T_e^{1/2} \alpha_1$	$T_e^{1/2} \alpha_B$	$T_e^{1/2} \beta_1$	$T_e^{1/2} \beta_B$	$T_e^{1/2} \beta_{ff}$	$T_e^{1/2} \beta_{ff}^{tot}$
1.0	1.646-11	9.283-11	1.646-11	8.287-11	1.061-11	9.348-11 *
1.2	1.646-11	8.823-11	1.646-11	7.821-11	1.068-11	8.889-11 *
1.4	1.646-11	8.361-11	1.646-11	7.356-11	1.076-11	8.432-11 *
1.6	1.646-11	7.898-11	1.646-11	6.892-11	1.085-11	7.977-11 *
1.8	1.646-11	7.435-11	1.646-11	6.430-11	1.095-11	7.525-11 *
2.0	1.646-11	6.973-11	1.645-11	5.971-11	1.106-11	7.077-11 *
2.2	1.645-11	6.512-11	1.644-11	5.515-11	1.118-11	6.633-11
2.4	1.645-11	6.054-11	1.643-11	5.062-11	1.132-11	6.194-11
2.6	1.644-11	5.599-11	1.641-11	4.614-11	1.145-11	5.758-11
2.8	1.642-11	5.147-11	1.638-11	4.170-11	1.161-11	5.332-11
3.0	1.640-11	4.700-11	1.633-11	3.734-11	1.181-11	4.915-11
3.2	1.636-11	4.258-11	1.625-11	3.306-11	1.202-11	4.508-11
3.4	1.629-11	3.823-11	1.613-11	2.888-11	1.224-11	4.112-11
3.6	1.620-11	3.397-11	1.594-11	2.484-11	1.248-11	3.733-11
3.8	1.605-11	2.983-11	1.565-11	2.098-11	1.274-11	3.373-11
4.0	1.582-11	2.584-11	1.522-11	1.736-11	1.303-11	3.039-11
4.2	1.548-11	2.204-11	1.460-11	1.402-11	1.335-11	2.737-11
4.4	1.499-11	1.847-11	1.374-11	1.103-11	1.369-11	2.472-11
4.6	1.431-11	1.520-11	1.260-11	8.442-12	1.403-11	2.247-11
4.8	1.341-11	1.226-11	1.119-11	6.279-12	1.434-11	2.062-11
5.0	1.227-11	9.696-12	9.571-12	4.539-12	1.460-11	1.914-11
5.2	1.093-11	7.514-12	7.844-12	3.192-12	1.478-11	1.797-11
5.4	9.454-12	5.710-12	6.146-12	2.185-12	1.485-11	1.704-11
5.6	7.920-12	4.257-12	4.601-12	1.458-12	1.482-11	1.628-11
5.8	6.427-12	3.117-12	3.295-12	9.484-13	1.468-11	1.563-11
6.0	5.058-12	2.244-12	2.262-12	6.023-13	1.444-11	1.505-11
6.2	3.866-12	1.590-12	1.494-12	3.738-13	1.414-11	1.451-11
6.4	2.877-12	1.110-12	9.520-13	2.268-13	1.380-11	1.402-11
6.6	2.089-12	7.642-13	5.878-13	1.348-13	1.344-11	1.358-11
6.8	1.485-12	5.199-13	3.528-13	7.859-14	1.311-11	1.318-11
7.0	1.036-12	3.498-13	2.066-13	4.499-14	1.280-11	1.285-11

* β_{ff} obtained by extrapolation and therefore slightly less accurate.

of 1 per cent. The summation of β_n behaves like that of α_n , but converges slightly more rapidly.

4 RESULTS AND DISCUSSION

Table 1 contains $T_e^{1/2}\alpha_1$, $T_e^{1/2}\alpha_B$, $T_e^{1/2}\beta_1$, $T_e^{1/2}\beta_B$, $T_e^{1/2}\beta_{ff}$ and $T_e^{1/2}\beta_B^{\text{tot}}$, the sum of the two previous quantities, for

$Z=1$. These quantities also appear in Fig. 2. All α s and β s scale with charge according to

$$\alpha(T_e, Z) = Z\alpha(T_e/Z^2, 1). \quad (10)$$

The ratios β_A/α_A and β_B/α_B are very close to 0.9 at $T_e = 10$ and decrease slowly to 0.13 and 0.18, respectively, at $T_e = 10^7$. The energy loss per recombination is thus always less than kT_e , significantly so at high temperatures.

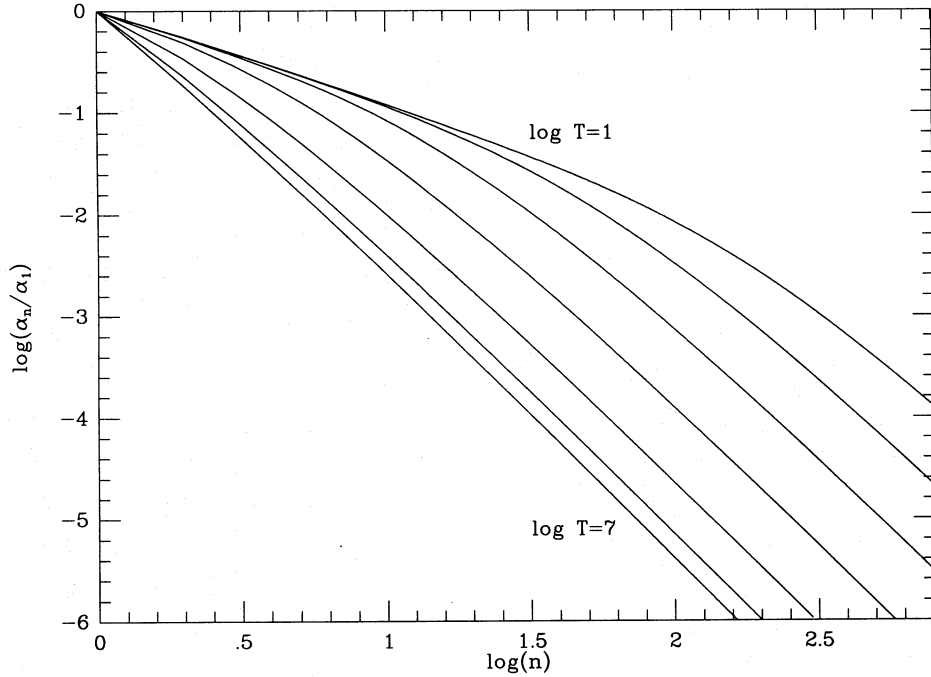


Figure 1. Logarithm of the ratio of recombination coefficients α_n to α_1 for $\log T_e = 1(1)7$. The transition from the approximately n^{-1} behaviour to the asymptotic form n^{-3} occurs at large n as T_e decreases.

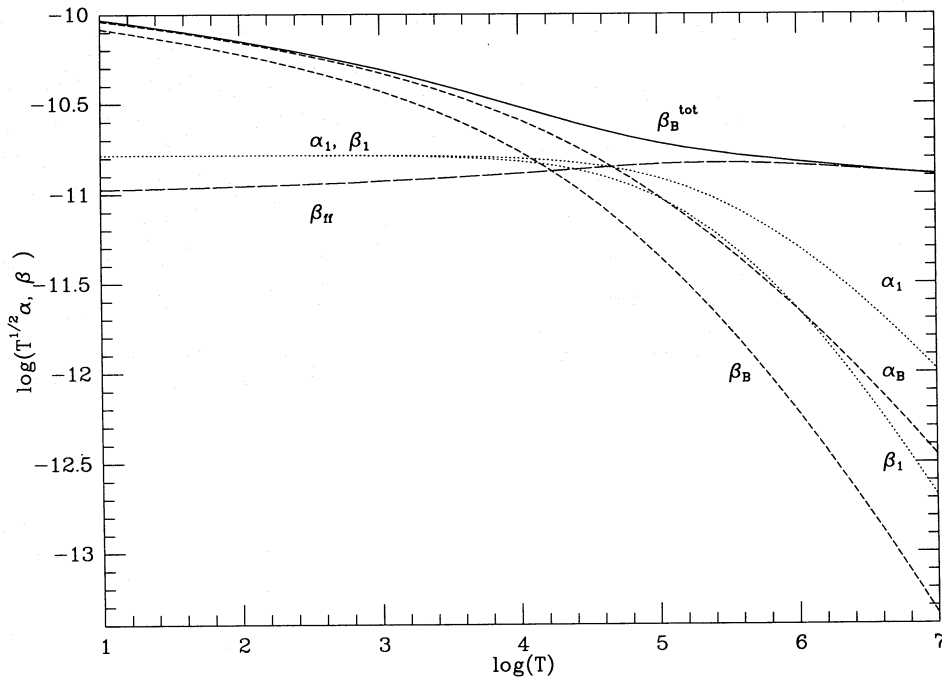


Figure 2. The temperature dependence of the recombination and energy-loss coefficients multiplied by $T_e^{1/2}$. Curves are labelled by the appropriate α or β .

Table 2. The ratios $\alpha_A(N_e)/\alpha_A(N_e=0)$ at selected temperatures.

$\log N_e$	$T_e = 5.10^2$	$T_e = 1.10^3$	$T_e = 1.10^4$	$T_e = 3.10^4$
-2	1.003			
-1	1.005			
0	1.009			
1	1.019			
2	1.036	1.016	1.001	1.000
3	1.068	1.032	1.002	1.000
4	1.126	1.060	1.004	1.001
5	1.236	1.112	1.008	1.003
6	1.450	1.209	1.016	1.004
7	1.893	1.398	1.031	1.008
8	2.908	1.783	1.055	1.015
9	5.635	2.647	1.109	1.029
10	14.95	4.896	1.205	1.056

Osterbrock (1989) gives $T_e^{1/2}\beta_H \approx 1.34 \times 10^{-11}$ for $T_e \approx 10^{-4}$, in good agreement with Table 1. At low temperatures Osterbrock's value is about 25 per cent too high, and at high temperatures it is about 10 per cent too low.

The present results are, strictly speaking, valid only for zero density. The ratio of $\alpha_A(N_e)$ to its value at zero density for several temperatures is given in Table 2, which is based on data kindly made available by P. J. Storey from the work of Storey & Hummer (1994) and from additional calculations. That the density dependence at low temperatures is very much stronger than at higher values can be understood in the following way. At low temperatures, for which the sums over n converge very slowly, the summations are strongly dominated by the terms for very large n . The populations of these levels are sensitive to collisional processes among themselves and with the continuum because of the small energy separations involved. As Fig. 1 shows, the higher levels become less and less important as the temperature increases.

The results in Table 1 were systematically compared with the values for α_1 , α_B , β_1 and β_B calculated from the tables and formulae given by Seaton (1959), which are based on a three-term asymptotic expansion of the Gaunt factor g_H (Burgess 1958). As expected, for $Z=1$ and $T_e \leq 10^4$, Seaton's results are quite accurate, with errors between 0.1 and 2.4 per cent. In the interval $10^4 \leq T_e \leq 10^6$, errors lie between 1 and 5 per cent, and at higher temperatures errors of 3 to 30 per cent are found. Seaton's simple formulae for α_A and β_A , given in equations (36) and (52) of his paper, are accurate to

better than 1.1 per cent for $\log T_e \leq 4.6$, above which the accuracy rapidly deteriorates.

Linear interpolation in the *logarithm* of the tabulated quantities is accurate in the worst case to a few units in the fourth significant figure, as is direct interpolation using a third-order Lagrange formula. Direct linear interpolation is accurate to a few tens of a per cent for all quantities for temperatures below roughly 20 000 K and for β_H and β_B^{tot} at all temperatures, while at the highest temperatures the errors are of order a few per cent. These interpolation procedures were evaluated by using them to interpolate to the mid-points of Table 1, and then again back to the original tabular points. As both interpolations were applied to essentially the same function and with the same mesh size, the error found by comparing the doubly interpolated values with the original is approximately twice the true error. This process appears to be much more reliable than the usual one of interpolating every other point back to the intermediate points, especially when high-order formulae are used. Because linear interpolation in a small, evenly spaced table (of logarithmic values) is so fast and easy to program, the original intention of making analytic fits to the data was not carried out.

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